



Decision Support

Stochastic model for physician staffing and scheduling in emergency departments with multiple treatment stages

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ABSTRACT

We propose a new solution for the Emergency Department (ED) staffing and scheduling problem, considering uncertainty in patient arrival patterns, multiple treatment stages, and resource capacity. A two-stage stochastic mathematical programming model was developed. We employed a Sample Average Approximation (SAA) method to generate scenarios and a discrete event simulation to evaluate the results. The model was applied in a large hospital, with 72,988 medical encounters and 85 physicians in a ten-month period. Compared to the hospital's actual scheduling, we obtained an overall average waiting time reduction from 54.6 (54.0–55.1) to 16.8 (16.7–17.0) minutes and an average Length of Stay reduction from 102.1 (101.7–102.4) to 64.3 (64.2–64.5) minutes. Therefore, this study offers a stochastic model that effectively addresses uncertainties in EDs, aligning physician schedules with patient arrivals and potentially improving the quality of service by reducing waiting times.

1. Introduction

Emergency departments (EDs) are the front line of acute healthcare defense in hospitals, where patients arrive with various illnesses and injuries. They serve as a center for medical treatment and a pathway for inpatient admissions (EL-Rifai, Garaix, Augusto, & Xie, 2015; Erhard, Schoenfelder, Fügner, & Brunner, 2018; Ganguly, Lawrence, & Prather, 2014). According to Wang et al. (2007), the demand for emergency care is increasing, and EDs may face dissatisfaction with long waiting times and length of stay. A typical response to these events is to increase capacity in the EDs, but it generates higher average unit costs and may improve the performance temporarily (Baboolal et al., 2012). To meet demand and control growing costs of service provision, healthcare administrators often need help to use resources as efficiently as possible without compromising patient satisfaction (Turhan & Bilgen, 2017). Mathematical techniques, including operations research, offer suitable tools for tackling modern healthcare challenges such as process design, capacity allocation, admission control, appointment, and staff scheduling (Barz & Rajaram, 2015; Peres, Hamacher, Cyrino Oliveira, Barbosa, & Viegas, 2019).

Satisfying personnel demands in the healthcare sector is a remarkably complex task due to the multi-skill nature of human resources. In effect, demand in healthcare systems must be addressed by appropriately qualified personnel and cannot be arbitrarily distributed between the resources (EL-Rifai et al., 2015).

The two typical phases of the staff allocation problem addressed are staffing and scheduling. The first determines the number of personnel in each shift to meet the demand, while the second assigns individual physicians to the shifts to meet the required staffing levels at different times. Staff scheduling at EDs is very challenging due to uncertainties (e.g., non-deterministic patient arrival times and service duration) and the complex system (e.g., different categories of patients, non-stationary arrival rates, and reentrant flow to the several “service stations”) (Kuo, 2014).

This paper proposes a generalized and integrated model to solve the staff scheduling problem, considering uncertainty in patient arrival patterns. The model considers staffing and scheduling for different time frames. We propose a two-stage stochastic mixed integer linear programming model to address uncertainty using the Sample Average Approximation (SAA) method and Latin Hypercube Sampling (LHS). The model includes different settings and considers physicians individually to account for their preferences and restrictions. From a practical perspective, this study identifies solutions to a relevant healthcare problem with stochastic aspects. The proposed model offers a valuable tool for healthcare practitioners to improve the efficiency and effectiveness of emergency department operations. We proposed a model to be applied in the tactical/operational level of resource capacity planning, considering the hierarchical framework for healthcare planning and control established by Hans, Van Houdenhoven, and Hulshof (2011).

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The structure of this paper is presented as follows. In Section 2, we outline the novel contributions of this study in the context of previous research. Section 3 comprehensively explains the problem formulation, while Section 4 details the proposed solution framework. Section 5 presents the results of the numerical experiment and a validating case study that confirms the effectiveness of the proposed methodology. Finally, in Section 6, we offer concluding remarks for the paper.

2. Literature review

This literature review analyses previous studies on physician staffing and scheduling problems, with a primary focus on work applied to medical emergencies and/or those considering uncertainty in the proposed methodology.

Regarding the scheduling phase for physicians in an ED, various approaches have been explored. Beaulieu, Ferland, Gendron, and Michelon (2000) devised a mathematical programming strategy using a commercial solver to generate schedules. Similarly, Carter and Lapierre (2001) conducted an analysis of schedules implemented across various hospitals. Ferrand, Magazine, Rao, and Glass (2011) applied integer programming to construct cyclic schedules. On the other hand, Lo and Lin (2011) put forward a Particle Swarm Optimization (PSO) method to optimize shift assignments for ED physicians. Additionally, Al Ghathbar, Louly, and Mrad (2019) employed an iterative local exploration method to address the challenge of arranging physician timetables, while Tan, Gan, Ren, et al. (2019) introduced a model utilizing mixed integer programming to organize physician groups as scheduling entities. Topaloglu (2006) presented a goal-oriented programming model for the scheduling of emergency medicine residents, taking into account both rigid and flexible constraints, the relative importance of which was determined through an analytical hierarchy process.

Some studies focus solely on determining the appropriate number of staff needed during specific time intervals without addressing the scheduling phase. For example, Green, Soares, Giglio, and Green (2006) and Jennings, Mandelbaum, Massey, and Whitt (1996) use queuing theory to calculate the proper staffing levels in EDs to reduce the number of non-served patients and to accommodate projected patient loads. Rossetti, Trzcinski, and Syverud (1999) proposed a simulation model to assess staffing levels' impact on the ED system's performance. Zeltyn et al. (2011) applied a simulation model to staffing scheduling problems across various time horizons. Izady and Worthington (2012) addressed the staffing and scheduling phases but in a non-integrated manner. They used an iterative approach combining a queuing model and simulation to determine the required number of medical staff. Peng, Yang, Strome, Weldon, and Chochinov (2020a, 2020b) proposed a quantitative approach using computer simulation to explore trade-offs between the efficiency, workload, and capacity of EDs. They applied alternative strategies to reduce patient waiting time and length of stay in COVID-19 scenarios.

Uncertainty is inherent in emergency care, and some works in the field of staffing and scheduling in EDs address the uncertainty in patient arrivals using different approaches. Hung, Whitehouse, O'Neill, Gray, and Kissoon (2007) developed a Physician Scheduling Analysis Tool (PSAT) using discrete-event simulation to test staffing scenarios for Pediatric Emergency Departments. Vile, Gillard, Harper, and Knight (2016) developed a Decision Support System (DSS) that predicts future demand using Singular Spectrum Analysis, recommends minimum staffing requirements using queuing theory, and generates low-cost schedules using linear programming. Kuo (2014) proposed a simulation–optimization approach that iteratively explores solutions to the problem. Savage, Woolford, Weaver, and Wood (2015) put forth a mixed integer programming model for optimizing ED physician staffing, leveraging historical patient arrival data. Devesse, Akartunali, Arantes, and Toledo (2022) proposed a reformulated mixed-integer programming model based on literature instances to tackle fairness in physician scheduling in EDs. Their approach provides balanced

schedules concerning target hours and weekends regarding fairness and high-demand coverage but does not include the uncertainty of patient arrival. Bozkir et al. (2023) proposed a two-stage model for the optimal allocation of dialysis machines during the pandemic. They considered uncertain patient demand scenarios but assumed known demands at the start of the day without incorporating uncertainty in patient arrivals throughout the day. Kraul and Brunner (2023) presented a two-stage stochastic formulation for residents' tactical planning with duty and daily scheduling's operational level. The authors considered the uncertainty in the absences of residents over a year but did not include the uncertainty in patient arrival.

Some papers address staffing and scheduling of different staff types (physicians, nurses, and technicians) and consider some uncertainty. Sinreich and Jabali (2007) introduced an iterative heuristic algorithm to develop weekday work shift schedules by adjusting shifts' start times to better correlate with ED resource demand. Sinreich, Jabali, and Dellaert (2012) proposed heuristic algorithms combining simulation and optimization models to schedule the work shifts of ED resources. Ganguly et al. (2014) estimated the workload demand distribution and proposed a chance-constrained model that assigns providers to admissible shifts to meet service levels and minimize costs. EL-Rifai et al. (2015) used a stochastic mixed-integer programming model solved with a Sample Average Approximation (SAA) method and evaluated resulting personnel schedules using a discrete-event simulation model. Still, they established the staffing level for each category and did not perform the scheduling. Sir et al. (2017) propose a method to manage patient arrival unpredictability in emergency departments (EDs). It uses classification and regression trees to set patient-to-staff ratio thresholds. The key point is the application of these thresholds and empirical cumulative distribution functions to historical patient volumes, which allows the determination of an optimal patient volume that accurately mirrors the ED's workload, considering the arrival patterns of ED patients. Çağlayan, Liu, Ayer, Pasupathy, Nestler, and Sir (2019) tackle the issue of unpredictable patient arrival times by creating a queuing network model. This model accurately represents the fluctuating flow of patients and distinguishes between patient–physician interactions and other services. They propose a staffing rule to maintain a steady tail probability of delay in emergency room treatment queues. However, their focus is solely on staffing, not scheduling. Marchesi, Hamacher, and Fleck (2020) included uncertainty in patient arrival to the emergency department and integrated the physician staffing and scheduling problem but focused specifically on staffing and scheduling physicians who work in the first assessment stage, i.e., only the door-to-doctor stage of an emergency. Prabhu, Taaffe, Pirrallo, Jackson, and Ramsay (2021) developed a deterministic physician optimization model and tested the proposed shifts in a simulation model with multiple ED stages, including the uncertainty in patient arrival and service times. The optimization model considered just the physician step (not comprising the multiple ED stages), and the shifts were the same for all days. Prabhu, Taaffe, Pirrallo, Jackson, and Ramsay (2022) propose a method that involves overlapping shifts and implementing restrictions on accepting new patients toward the end of a physician's shift. This approach is designed to manage the inherent uncertainty in patient arrival and service times. To assess the impact of these policies on ED performance metrics, the study employs a hybrid simulation model that takes into account the aforementioned uncertainties. Similar to the study by Çağlayan et al. (2019), the focus in Prabhu et al. (2022) study is exclusively on staffing considerations rather than on scheduling.

Another highly relevant aspect to consider in physician staffing and scheduling is the integration of breaks into the models. This feature is crucial for maintaining workforce efficiency and satisfaction but is often overlooked in traditional scheduling approaches. Given that this aspect is particularly relevant to this research, which includes breaks in its modeling to enhance the realism and applicability of the scheduling solution, it is worthwhile to cite the works of Brunner, Bard, and Kolisch (2009, 2011) and Stolletz and Brunner (2012). Even though

these articles do not apply directly to the study's focus on medical emergencies, they collectively highlight the importance of integrating breaks into scheduling models. One more common theme across the three articles is the development of flexible shift scheduling models for physicians, emphasizing adaptability in shift start times, lengths, and break placements without adhering to predefined shift patterns.

Table 1 compares literature on physician staffing, considering planning, shift, and personnel. This paper is also included to position it within the context of previous research (see column explanations in Supplementary Material S1).

From Table 1, regarding the uncertainty and techniques used by each article, we can summarize the main applied approaches to better understand the current literature on ED scheduling and staffing. There were different approaches to include uncertainty, which could be a stochastic optimization model, a Discrete Event Simulation Model (DES) or a Queuing Model (QM). The uncertainty in the arrival process and service time were mostly included in articles with DES or QM approaches, since it is simpler and natural to include them in simulation or queueing models (QM: (Çağlayan et al., 2019; Green et al., 2006; Jennings et al., 1996); DES: (Hung et al., 2007; Peng et al., 2020a, 2020b; Prabhu et al., 2022; Rossetti et al., 1999; Zeltyn et al., 2011)). Most stochastic optimization articles were capable of including the uncertainty in the arrival process (Daldoul, Nouaouri, Bouchriha, & Allaoui, 2018; Ganguly et al., 2014; Vile et al., 2016) and some of them used DES models to include the uncertainty in the service time (EL-Rifai et al., 2015; Kuo, 2014; Prabhu et al., 2021).

In the domain of healthcare staff scheduling in medical emergencies, this paper offers a new approach by integrating comprehensive methodologies and addressing critical aspects overlooked by others. While EL-Rifai et al. (2015), Kuo (2014), Prabhu et al. (2021) share a similar focus on stochastic modeling and discrete event simulation, they fall short in integrating staffing and scheduling with a multistage treatment perspective. Notably, their planning horizon is limited to only one day, and they do not consider break times and preferences. Our paper, however, offers a flexible planning horizon that can be redefined according to the users' needs and adeptly fills this gap by addressing both cyclic and acyclic shifts, as well as incorporating considerations for break times and physician preferences. Moreover, this study advances beyond the contributions of Peng et al. (2020a, 2020b) by utilizing discrete event simulation and embedding an approach to uncertainty that accounts for the unpredictable nature of emergency department operations.

Additionally, while Prabhu et al. (2021, 2022) incorporate mixed integer programming and discrete event simulation, they did not address shift breaks and physician preferences. Similarly, Zeltyn et al. (2011), despite focusing on simulation for scheduling challenges, does not encapsulate the integrated approach or handle uncertainty as this paper does. It also does not address break times, multiple stages, and focuses solely on the staffing phase. Our study distinguishes itself from Kraul and Brunner (2023) by applying stochastic programming in conjunction with discrete event simulation, offering a more robust framework for understanding and planning around healthcare uncertainties. Compared to Brunner et al. (2009), Brunner and Edenharter (2011) and Stolletz and Brunner (2012), our work focuses on the emergency department and does not adopt the flexible scheduling strategy. Instead, it allocates physicians to shifts predefined by hospital administrators, prioritizing administrative rules over flexibility.

In conclusion, as revealed in Table 1, our study significantly contributes to the existing literature by addressing staffing and scheduling in a unified manner, integrating multiple stages in the emergency department and considering breaks. Our research offers a flexible model for shift characteristics, including user-defined planning horizon, number of shifts, shift duration, and start times, accommodating physician preferences and allowing for both cyclic and acyclic shift periodicity. This adaptability, combined with our unique approach to incorporating intervals for shifts and addressing the complexities of emergency

department operations, makes our formulation distinct and capable of tackling real-world challenges in various hospital settings.

In this context, this article contributes to both conceptual and practical perspectives. From a conceptual perspective: (i) we propose a generalized and integrated model that considers uncertainty in patient arrival patterns and simultaneously accounts for ED physician staffing and scheduling problems; (ii) we consider several stages and resources of ED treatment, including bed/chair constraints and nurse capacity limitations; (iii) we include staffing and scheduling for different time frames by defining a planning horizon with flexibility; (iv) we consider uncertainty by hour and day of the week; (v) we suggest a two-stage stochastic approach using the Sample Average Approximation (SAA) method and Latin Hypercube Sampling (LHS); (vi) we include different settings to increase model flexibility and address more realistic applications (e.g. breaks consideration). We also consider individual physician preferences/restrictions. From a practical perspective, the study identifies solutions to a relevant healthcare problem, including stochastic aspects of the system.

3. Problem statement

3.1. Healthcare staff scheduling problem features definition

Marchesi et al. (2020) outlined the key features of the healthcare staff scheduling problem. These include the planning horizon, shift types, and personnel characteristics, which are explained as follow. The planning horizon refers to the period in which scheduling occurs. Shift types are defined in terms of periodicity (cyclic or acyclic shifts), overlay (overlapping or non-overlapping shifts), and restriction of start/end times (fixed or definable shifts). In cyclic shifts, employees in a specific category perform a recurring pattern or tour of the same type of work repeated in every planning period. On the other hand, acyclic shifts involve independent work assignments, and a new schedule is created for each planning period. Adjacent shifts that share a portion of the work period are called overlapping shifts, while those with no intersection of hours are known as non-overlapping shifts. Shifts with a clearly defined start, length, and end time are referred to as fixed shifts, while shifts where the start time, end time, and length are decision variables are known as definable shifts. In addition to shift types, it is essential to consider physicians' break intervals within their shifts. These breaks are crucial for rest and recuperation, ensuring sustained patient care quality and preventing staff fatigue. Finally, personnel characteristics refer to information on how many skill categories are considered and which scheduling rules must be observed for the different categories.

3.2. Detailed problem description

Fig. 1 illustrates the clinical pathways in an ED. First, the patients go to a nurse (triage) who determines whether they need vital medical care, leading to three possible alternatives: (i) Patients not requiring vital care but needing special conditions of treatment are directed to the "Observation Hold"; (ii) Patients in need of vital care are immediately taken to the "Life-Threatening Emergencies" area; (iii) Patients who do not require vital care or special treatment are placed in a waiting room until a physician is available to do the "First General Assessment". From the Life-Threatening Emergencies, patients may either move to "Observation Hold" for further monitoring or undergo "Auxiliary Exams". After the First General Assessment, patients categorized in case iii can proceed through various stages based on their medical needs. They may receive "Medication", undergo "Auxiliary Exams", be subject to "Subsequent patient evaluations", or be moved to "Observation Hold". It is important to note that the 'Subsequent patient evaluations' stage is designed to allow for reentries, enabling patients to undergo this assessment multiple times based on their evolving medical needs. This flexibility in the process is crucial in accommodating the dynamic nature of patient care.

Table 1
Comparative evaluation of this paper versus existing literature.

Paper (A)	PH (B)	Shift's number (C)				Shift overlapping (D)		Shift length (E)		Shift start (F)		Shift periodicity (G)		Skill categories (H)				SB (I)	PPR (J)	Problem type (K)			MS (L)	Uncertainty (M)	Technique (N)	
		2	3	+	UD	NO	O	F	D	F	D	C	AC	1	+	Ssep.	Hsub.			Staff.	Sched.	SSI				
Al Ghatbbar et al. (2019)	1 m		x			x		x		x			x	x	x			x		x				ILS		
Beaulieu et al. (2000)	3 m			x			x	x		x			x	x	x			x		x				IP; TS		
Carter and Lapierre (2001)	1y	x				x		x		x			x		x	x		x		x				MIP		
Çağlayan et al. (2019)	1d			x		x		x			x		x	x					x			x	x	QM		
Daldoul et al. (2018)	1d			x			x		x		x		x		x	x			x			x	x	SP		
Devesse et al. (2022)	UD				x			x		x			x	x				x		x				MIP		
El-Rifai et al. (2015)	1d			x			x			x		x	x		x	x			x			x	x	SP; DES		
Ferrand et al. (2011)	1y			x			x		x			x		x		x				x				IP		
Ganguly et al. (2014)	1d			x			x	x		x			x		x	x	x		x				x	CC		
Green et al. (2006)	1w			x			x				x		x	x	x			x		x			x	QM		
Hung et al. (2007)	1d			x			x	x			x		x	x	x				x				x	DES		
Izady and Worthington (2012)	1d			x			x				x		x		x	x			x	x		x		QM; Sim		
Jennings et al. (1996)	1d	na	na	na	na	na		na	na	na	na	na	na	x	x			na		x			x	QM		
Kraul and Brunner (2023)	1y	na	na	na	na	na		na	na	na	na	na	na	x				x		x	x		x	SP		
Kuo (2014)	1d			x			x	x		x			x	x	x				x				x	Sim-Opt		
Lo and Lin (2011)	UD			x			x			x			x	x	x			x		x				PSO		
Peng et al. (2020a)	6 m	na	na	na	na	na		na	na	na	na	na	na		x				x			x	x	DES		
Peng et al. (2020b)	6 m			x			x	x		x			x		x				x				x	DES		
Prabhu et al. (2021)	3w			x			x	x			x		x	x	x						x		x	MIP; DES		
Prabhu et al. (2022)	3w			x			x	x		x			x	x	x						x		x	DES		
Rossetti et al. (1999)	1w			x			x				x		x	x	x								x	DES		
Savage et al. (2015)	1d			x			x				x		x	x	x					x				MIP		
Sinreich and Jabali (2007)	1d			x				x	x			x			x	x				x		x	x	Sim-Opt		
Sinreich et al. (2012)	1d			x			x			x	x		x			x				x		x	x	Sim-Opt		
Sir et al. (2017)	1w			x			x			x		x		x	x	x		x			x		x	MIP; IDV; CART		
Tan et al. (2019)	1 m			x			x	x		x			x	x	x			x			x			MIP		
Topaloglu (2006)	1 m			x			x			x			x		x	x				x				GP		
Vile et al. (2016)	1 m			x			x	x		x			x	x	x				x	x			x	QM; IP		
Zeltny et al. (2011)	UD			x			x			x			x		x	x				x			x	DES		
This paper	UD				x		x	x		x		x	x	x	x			x	x			x	x	SP; DES		
Total	na	1	7	11	1	11		11	16	5	13	7	4	18	12	9	21	1	1	8	12	11	1	9	13	na

¹ PH — Planning horizon; w — week(s); m — month; d — day(s); y — year; UD — user-definable; +More shifts; F — fixed; D — definable; NO — non-overlapping; O — overlapping; C — cyclic; AC — acyclic; SB — Shifts' break; PPR — Physician preferences and/or restrictions; Staff. — Staffing; Sched. — Scheduling; SSI — Staffing and Scheduling Integrated; MS — Multiple Stages; IP — Integer Programming; TS — Tabu Search; MIP — Mixed Integer Programming; SP — Stochastic Programming; DES — Discrete Event Simulation; CC — Chance Constraint Model; ILS — Iterated Local Search; QM — Queueing Model; Sim — Simulation; Sim-Opt — Simulation-Optimization; PSO — Particle Swarm Optimization; GP — Goal Programming.

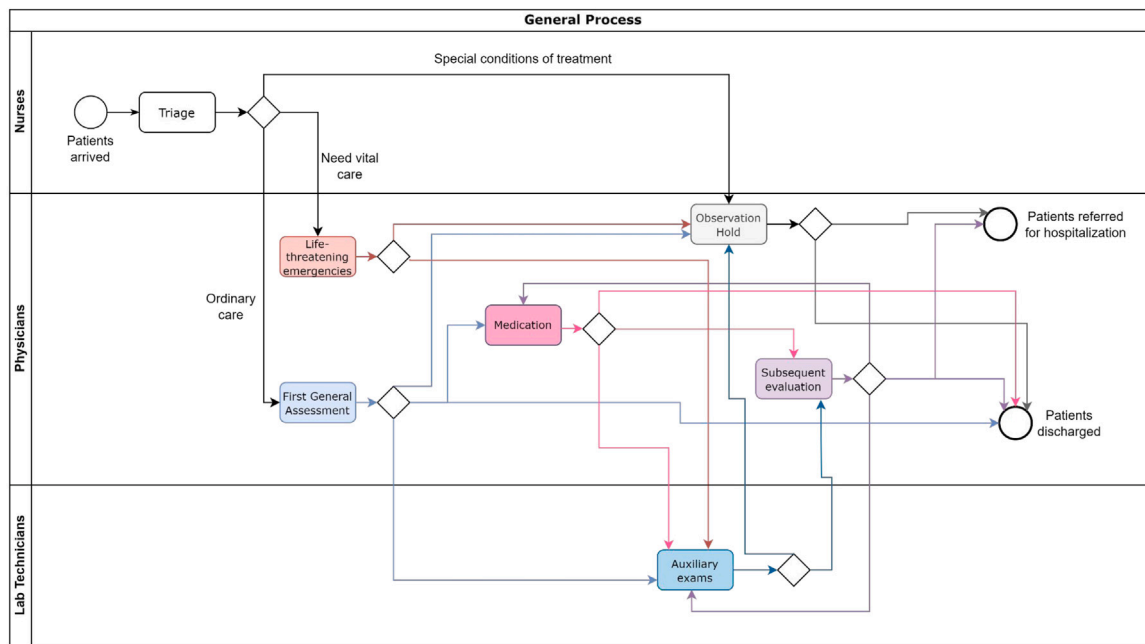


Fig. 1. BPMN notation of ED patient flow.

Physicians perform health assessments, examine patients in the medication phase, and examine patients in Life-Threatening room and Observation Hold; nurses handle the triage of patients; and lab technicians manage the auxiliary examinations. The latter are out of the scope of this work since they are shared with other departments. As [Daldoul et al. \(2018\)](#), we consider a limit for the critical resources, such as chairs in the medication phase and beds in the observation hold. Each activity represents a treatment stage in the ED. Regarding the applicability and generalization of the service process diagram, our process has similar characteristics to other ED processes reported by different studies ([Peng et al., 2020a, 2020b; Prabhu et al., 2021, 2022](#)). The “Life-Threatening emergency” is also reported in the literature as a first assessment for urgent patients. The triage, first assessment, medication, auxiliary exams, observation hold, and subsequent evaluations are everyday ED activities.

The problem presents physician scheduling and contractual rules related to (i) individual workload for the entire horizon, buffering between weekdays and weekends; (ii) availability and individual preferences for start time in the ED; (iii) maximum allowed consecutive hours of work; (iv) a maximum number of shifts per day; (v) minimum interval between allocations; and (vi) shift assignment rules (e.g., the physician can work in more than one shift per day; however, shifts must be non-overlapped and sequential; whether the physician is allocated to the night shift, the interval between assignments must be higher than a given value).

Our framework incorporates characteristics that make it compliant and capable of accommodating actual emergency department configurations, including (i) a scheduling horizon and shift's cycle frequency defined by the user, (ii) multiple overlapping shifts whose length and beginning can be altered, (iii) the possibility to include breaks, considering them as different shifts, (iv) different physician specialties; (v) patient arrival patterns that vary by hour and day, implying that patient arrival rates change throughout the day but also between days; (vi) break intervals in physicians' shifts.

4. Solution framework

We follow the solution framework composed of four steps proposed by [Marchesi et al. \(2020\)](#). Step 1 represents the data collection, and step 2 is the statistical analysis (both using real data). Step 3 composes

a two-stage stochastic optimization model — solved using the Sample Average Approximation (SAA) technique, and step 4 presents a discrete event simulation model (both steps using simulated data).

Data collection involves gathering information about processes, business rules, service requirements, and agreements in the hospital's emergency department. Statistical analysis determines the probability distribution of patient arrivals, used to generate scenarios with SAA and run the discrete event simulation model.

The fundamental concept of Sample Average Approximation (SAA), initially proposed by [Shapiro and Homem-de Mello \(1998\)](#), is to estimate the optimal solution value by calculating the average of the objective function over instances composed of subsets of scenarios sampled successively and independently. This approach approximates the optimal solution without directly evaluating the objective function over the entire scenario set. [Linderoth, Shapiro, and Wright \(2006\)](#) demonstrated that by employing the SAA methodology and building upon the concepts introduced by [Norkin, Pflug, and Ruszczyński \(1998\)](#) and [Mak, Morton, and Wood \(1999\)](#), it is feasible to derive both upper and lower bounds for the optimal value of the actual problem.

The process of integrating SAA into our numerical experiments is outlined in Algorithm 1, which can be found in Supplementary Material S2. This involves specifying several key parameters: M , representing the number of replications and the initial N for lower bound calculations; R , indicating the number of replications, and $\bar{N}$, representing the number of scenarios for upper bound calculations; ϵ , which serves as the stopping criterion; and δ , the increment value for adjusting the number of scenarios, N . The equations required for computing the variables in Algorithm 1 can be found in Supplementary Material S2. In our research, we selected ϵ to represent a 1% difference between lower and upper bounds, which translates to estimating a gap of 1%. In practical terms, this estimation considers a 95% confidence interval (CI), aiming for the precision of the CI to be within 2%. The determination of the required number of replications to achieve this level of precision follows the methodology established by [Hoad, Robinson, and Davies \(2010\)](#).

In this paper, the SAA sampling technique employed was the Latin Hypercube Sampling (LHS). [Firestone et al. \(1997\)](#) states that LHS is considered more efficient than MCS since it requires fewer simulations to produce the same level of precision. It is generally recommended when the model is complex or time and resource constraints are an

issue.

We proposed a two-stage stochastic optimization model that considers the seven stages of treatment in the emergency department (ED) and includes physician staffing, scheduling, bed/chair constraints, and nurse capacity limitations. We adapted the scenario generation and discrete event simulation stages proposed by Marchesi et al. (2020) to encompass the entire flow of the medical emergency.

4.1. Two-stage stochastic optimization model

The optimization model determines the daily requirements of nurses and physicians through the staffing step and assigns individual physicians to each shift via the scheduling step. These decisions, represented by integer and binary variables, are known as first-stage variables and are made before the realization of uncertainties. The model considers fluctuations in demand, described as possible scenarios of emergency department patient visits, as discrete sources of uncertainty. Second-stage decisions are made with full knowledge of the uncertainty, including factors such as the number of patients waiting in a queue and resource utilization levels.

The mathematical programming model is based on the problem described in the previous section and represented by Fig. 2. In this section, queue, and stages of treatment are used as synonyms. The number of patients arriving in the system on each period t and hour h is denoted DEM_{ht} . The variable w_{htq} counts the number of patients waiting for service q . The number of patients served depends on the transfer rates $(\alpha_q, \beta_q, \eta_q, \mu_q, \rho^1, \rho^2)$.

We made some assumptions in the mathematical model following the definitions made by Daldoul et al. (2018) and EL-Rifai et al. (2015) (some of these will be relaxed in the simulation model): (i) The system is initially empty. For this, we start the scheduling horizon at an hour when the system is usually empty (Daldoul et al., 2018; EL-Rifai et al., 2015); (ii) We consider the physicians from the same skill categories and the nurses to be homogeneous, i.e., with the same service time (EL-Rifai et al., 2015); and (iii) Auxiliary examinations and treatment in the life-threatening emergency room are assumed to have a fixed duration, and we do not consider staffing for these stages.

In our study, we start by modeling the emergency department from an empty state, aligning with low activity periods to provide a simplified baseline for our optimization model. This assumption is refined in the simulation phase with a warm-up period to mimic fluctuating occupancy. We treat service times for physicians and nurses within the same skill categories as homogeneous to streamline the model and focus on demand variability's impact on resource allocation. This is further elaborated in the simulation using probabilistic distributions to model service times. For auxiliary examinations and treatments, initially assumed as fixed durations based on hospital-wide averages, we use probabilistic distributions in the simulation to reflect variability and interconnectedness with overall hospital operations. Our model's initial simplifications are revisited and refined in the simulation phase, balancing analytical tractability with real-world emergency department dynamics.

The model focuses on creating a plan for a series of consecutive days, referred to as the D planning horizon. This horizon can be separated into T segments known as staffing periods, where staffing levels will be determined and repeated as defined. It is essential to highlight that the proposed model can handle both cyclic and acyclic staffing and scheduling scenarios, depending on how the parameters $PLAN_HORIZON$ and $PLAN_PERIOD$ are defined. For a comprehensive understanding of how these parameters influence shift periodicity determination, we refer the reader to Supplementary Material S3, which provides a detailed explanation. Tables 2–4 comprehensively presents sets, model parameters, and variable definitions.

The plan arranges a fixed number of physicians P from K diverse skill categories throughout the planning horizon. Each day within the horizon, with H hours, is divided into S shifts that may overlap.

The plan also accommodates break intervals within shifts, ensuring each physician has designated rest periods (more explanations also in Supplementary Material S3) The workload for each shift is defined by its start time and duration. Based on their clinical pathway, patients may undergo various stages Q during their clinical journey.

Our model can be adapted to reflect the operational characteristics of several EDs and consider other stages of treatment by making minor changes. For clarity in the notation, the summation domains are not included unless the summation is only evaluated on a subset of the natural domain. Without such mention, it should be assumed that the domain is the original set referred to by the index. The mathematical formulation proposed is:

$$\text{Min} Z = \sum_h \sum_t \sum_q \sum_c \text{PROB}_c w_{htq}^c \quad (1)$$

Subject to:

$$\begin{aligned} & \sum_{d \in WD} \sum_{s \in FS} x_{sdp} (S_END_s - S_START_s + 1) + \\ & \sum_{d \in WD} \sum_{s, s' \in FS} x_{sdp} (S_END_s - S_END_{s'} + 1) \\ & * COM_SHIFTS_{ss'} \leq WL_WEEK_p \end{aligned} \quad (2)$$

$$\forall p \in P1$$

$$\begin{aligned} & \sum_{d \in WD} \sum_{s \in FS} x_{sdp} (S_END_s - S_START_s + 1) + \\ & \sum_{d \in WD} \sum_{s, s' \in FS} x_{sdp} (S_END_s - S_END_{s'} + 1) \\ & * COM_SHIFTS_{ss'} \leq WL_WEEKEND_p \end{aligned} \quad (3)$$

$$\forall p \in P2$$

$$\begin{aligned} & \sum_d \sum_{s \in FS} x_{sdp} (S_END_s - S_START_s + 1) + \\ & \sum_d \sum_{s, s' \in FS} x_{sdp} (S_END_s - S_END_{s'} + 1) \\ & * COM_SHIFTS_{ss'} \leq WL_TOTAL_p \end{aligned} \quad (4)$$

$$\forall p \in P$$

$$\begin{aligned} & x_{sdp} + x_{s'dp} * COM_SHIFTS_{ss'} \geq 2 \\ & \forall p \in PP, d \in D, s \in S, s' \in S \mid s \neq s' \end{aligned} \quad (5)$$

$$\sum_s x_{sdp} \leq MAX_ASSIGN^1 \quad (6)$$

$$\forall p \in PP, d \in D$$

$$\begin{aligned} & x_{sdp} + x_{s'dp} (1 - REL_SHIFTS_{ss'}) \leq MAX_ASSIGN^1 \\ & \forall p \in PP, d \in D, s \in YS, s' \in S \mid s \neq s' \end{aligned} \quad (7)$$

$$x_{sdp} + \sum_{s' \mid s' \neq s} x_{s'dp} REL_SHIFTS_{ss'} \leq MAX_ASSIGN^2 \quad (8)$$

$$\forall p \in PP, d \in D, s \in YS$$

$$\sum_{s \in Night} x_{sdp} + x_{s',d+1,p} \leq MAX_ASSIGN^1 \quad (9)$$

$$\forall p \in PP, d \in D, s' \in S$$

$$\sum_d x_{sdp} \leq MAX_TYPE_{ps} \quad (10)$$

$$\forall p \in PP, s \in S \mid REL_PHY_{ps}$$

$$\sum_{p \in P_k} \sum_d \sum_s \sum_{h'} x_{sdp} START_COMB_{sh'p} \geq MIN_{kh} \quad (11)$$

$$\forall h \in H, k \in K$$

$$\sum_{p \in PP} \sum_d \sum_s \sum_{h'} x_{sdp} START_COMB_{sh'p} \geq \sum_q nPhy_{htq} \quad (12)$$

$$\forall h \in H, t \in T \mid DAY_PERIOD_{dt}$$

$$s_{htq}^c \leq nNur_{ht} CAP_{htq} \quad (13)$$

$$\forall h \in H, t \in T, q \in Q \mid q = 1, c \in SC, p \in NP$$

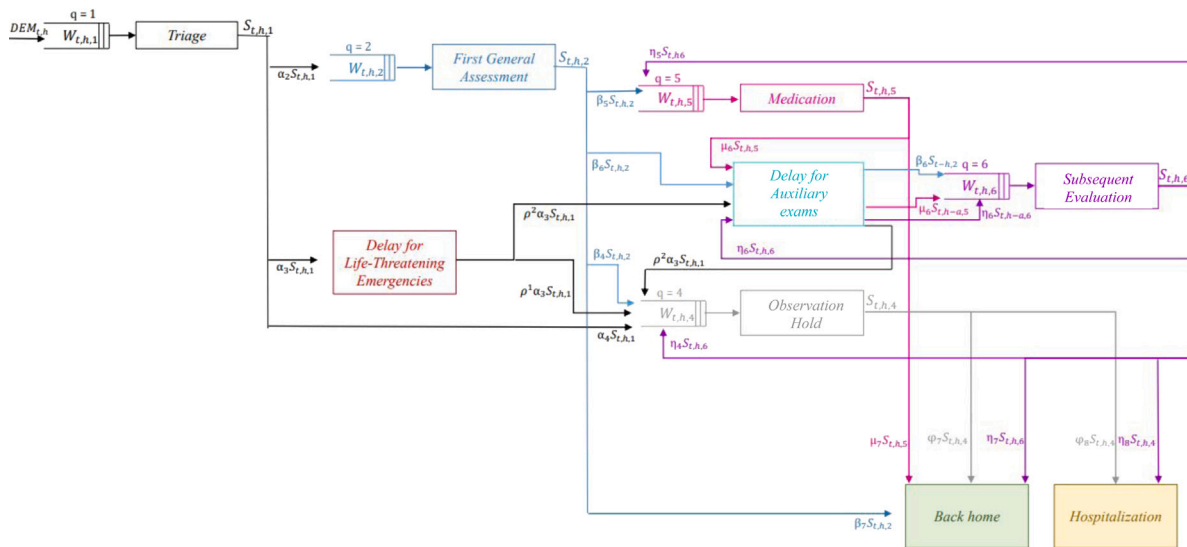


Fig. 2. ED queue model.

Table 2

Sets, subsets and corresponding domains of the general model.

Sets	Indexes	Domain	Description
Shifts (S)	s, s'	$\{1, \dots, S \}$	Shifts in the ED
Staff (P)	p	$\{1, \dots, P \}$	Physicians and nurses in the ED
Days (D)	d	$\{1, \dots, PLAN_HORIZON\}$	Days of the planning horizon
Hours (H)	h, h'	$\{0, \dots, 23\}$	Hours of the day (24 h clock format)
Period (T)	t, t'	$\{1, \dots, PLAN_PERIOD\}$	Period over which staffing level is repeated
Physician Category (K)	k	$\{1, \dots, K \}$	Physician skill categories
Stages of treatment (queue) (Q)	q	$\{1, \dots, Q \}$	Stages of treatment (queue) in the ED
Scenarios (SC)	c	$\{1, \dots, SC \}$	Possible realization scenarios
Staff in certain category (P_k)	–	Subset of Staff (P)	Physicians who belong to category k
Nurse Staff (NP)	–	Subset of Staff (P)	Staff who is Nurse
Physician Staff (PP)	–	Subset of Staff (P)	Staff who is Physician
Weekdays (WD)	–	Subset of Days (D)	Weekdays on the planning horizon
Weekends (W)	–	Subset of Days (D)	Weekend days on the planning horizon
Night Shifts ($NIGHT$)	–	Subset of Shifts (S)	Night shifts
N_Shifts (NS)	–	Subset of Shifts (S)	Shifts that do not allow combined allocation
Y_Shifts (YS)	–	Subset of Shifts (S)	Shifts that allow combined allocation
F_Shifts (FS)	–	Subset of Shifts (S)	Shifts that are the first in the sequence of shifts that need to be combined due to the interval.
Physicians_Week ($P1$)	–	Subset of Physicians (P)	Physicians that work only weekdays
Physicians_Both ($P2$)	–	Subset of Physicians (P)	Physicians that work both weekdays and weekends

Table 3

General model variables.

Variables	Description	Domain
w_{htq}^c	Number of patients waiting in each hour h , each period t , each queue q and each scenario c	$\mathbb{R}_+$
x_{sdp}	Decision if physician p is assigned in the shift s in the day d	$\{0, 1\}$
$nPhy_{htq}$	Number of physicians required for each hour h , each period t and stage of treatment q	$\mathbb{Z}_+$
s_{htq}^c	Number of patients served in each hour h , each period t , each stage of treatment q and each scenario c	$\mathbb{R}_+$
$nNur_{ht}$	Number of nurses required for each hour h , and each period t	$\mathbb{Z}_+$

Table 4
General model parameters.

Parameters	Description	Unit
$PROB_c$	Probability of each scenario c	–
$PLAN_HORIZON$	Number of days in the planning horizon	–
$PLAN_PERIOD$	Period (number of days) over which the staffing level is repeated	–
S_START_s	Start hour of each shift s	Hour
S_END_s	End hour of each shift s	Hour
WL_WEEK_p	Workload available for each staff p in the weekdays	Hour
$WL_WEEKEND_p$	Workload available for each staff p in the weekends	Hour
WL_TOTAL_p	Total workload available for each staff p	Hour
MAX_ASSIGN^1	Maximum number of assignments per day to shift that does not allow combined allocation	–
$REL_SHIFTS_{ss'}$	Indication if shift s can be combined with shift s'	{0, 1}
$COM_SHIFTS_{ss'}$	Indication if shift s need to be combined with shift s'	{0, 1}
MAX_ASSIGN^2	Maximum number of assignments per day to shift that allows combined allocation	–
MAX_TYPE_{ps}	Maximum number of assignments for staff p in a certain shift type s	Hour
REL_PHY_{ps}	Indication if staff p can be scheduled for shift s	{0, 1}
$START_COMB_{shp}$	Indication of start time h of each shift s for each staff p	{0, 1}
MIN_{kh}	Minimum number of staff required for each skill category k in each hour h	–
DAY_PERIOD_{dt}	Indication if day d is included in period t	{0, 1}
DEM_{ht}^c	Demand (arrival of patients for each period t , hour h , and scenario c)	–
α_q	Transfer rate of patients from the Triage to the stage of treatment q	[0,1]
β_q	Transfer rate of patients from the First General Assessment to the stage of treatment q	[0,1]
η_q	Transfer rate of patients from the subsequent evaluation to the stage of treatment q	[0,1]
μ_q	Transfer rate of patients from the Medication to the stage of treatment q	[0,1]
ρ^1	Transfer rate of patients from the Life-Threatening room direct to Observation Hold	[0,1]
ρ^2	Transfer rate of patients from the Life-Threatening room to auxiliary examinations	[0,1]
MAX_R_{ht}	Maximum number of registration staff available per hour h and day t	–
MAX_PHY_{htq}	Minimum number of staff required per hour h , day t and stage of treatment q	–
NC	Number of chairs available to treatment in Medications step	–
NB	Number of beds available to treatment in Observation Hold room	–
CAP_{htqp}	Capacity (service rate) per hour h , day t , stage of treatment q , staff p	–
$I_PHYSICIAN_{sdp}$	Indication if physician p is available on day d and shift s	{0, 1}

$$s_{htq}^c \leq nPhy_{htq} CAP_{htqp} \quad \forall h \in H, t \in T, q \in Q \mid (q = 2 \vee q = 4 \vee q = 5 \vee q = 6), c \in SC, p \in PP \quad (14)$$

$$s_{htq}^c \leq w_{htq}^c \quad \forall h \in H, t \in T, q \in Q, c \in SC \quad (15)$$

$$s_{htq}^c \leq NC \quad \forall h \in H, t \in T, q \mid q = 5, c \in SC \quad (16)$$

$$s_{htq}^c \leq NB \quad \forall h \in H, t \in T, q \mid q = 4, c \in SC \quad (17)$$

$$nNur_{ht} \leq MAX_R_{ht} \quad \forall h \in H, t \in T \quad (18)$$

$$nPhy_{htq} \geq MIN_PHY_{htq} \quad \forall h \in H \mid h \geq 1, t \in T, q \in Q \quad (19)$$

Triage Queue

$$w_{ht'q}^c = w_{h-1,t',q}^c + DEM_{htq}^c - s_{h-1,t',q}^c \quad \forall h \in H, t \in T, t' \in T, q \in Q \mid q = 1, c \in SC \quad (20)$$

First General Assessment Queue

$$w_{htq}^c = w_{h-1,t',q}^c + \alpha_q s_{h-1,t',q-1}^c - s_{h-1,t',q}^c \quad \forall h \in H, t \in T, t' \in T, q \in Q \mid q = 2, c \in SC \quad (21)$$

Medication Queue

$$w_{htq}^c = w_{h-1,t',q}^c + \beta_q s_{h-1,t',q-3}^c + \eta_q s_{h-1,t',q+1}^c - s_{h-1,t',q}^c \quad \forall h \in H, t \in T, t' \in T, q \in Q \mid q = 5, c \in SC \quad (22)$$

Second General Assessment Queue

$$w_{htq}^c = w_{h-1,t',q}^c + \beta_q s_{h-2,t'',q-4}^c + \mu_q s_{h-2,t'',q-1}^c + \eta_q s_{h-2,t'',q}^c - s_{h-1,t',q}^c \quad \forall h \in H, t \in T, t' \in T, q \in Q \mid q = 6, c \in SC \quad (23)$$

Beds Queue

$$w_{htq}^c = w_{h-1,t',q}^c + \alpha_q s_{h-1,t',q-3}^c + \beta_q s_{h-1,t',q-2}^c + \alpha_{q-1} * \rho^1 s_{h-1,t'',q-3}^c + \alpha_{q-1} * \rho^2 * s_{h-1,t''',q-3}^c + \eta_q s_{h-1,t',q+2}^c - s_{h-1,t',q}^c \quad \forall h \in H, t \in T, t' \in T, q \in Q \mid q = 4, c \in SC \quad (24)$$

$$w_{htq}^c \in \mathbb{R}_+, \quad \forall h \in H, t \in T, q \in Q, c \in SC \quad (25)$$

$$x_{sdp} \in \{0, 1\}, \quad \forall s \in S, d \in D, p \in PP \mid \exists I_PHYSICIAN_{sdp} \quad (26)$$

$$nPhy_{htq} \in \mathbb{Z}_+, \quad \forall h \in H, t \in T, q \in Q \quad (27)$$

$$s_{htq}^c \in \mathbb{R}_+, \quad \forall h \in H, t \in T, q \in Q, c \in SC \quad (28)$$

$$nNur_{htq} \in \mathbb{Z}_+, \quad \forall h \in H, t \in T, q \in Q \quad (29)$$

The objective function in Eqs. (1) aims to minimize the expected total number of patients waiting in the queue, which is calculated as the sum of patients waiting per hour h and period t , across all scenarios c (w_{htq}^c), weighted by the probability of each scenario ($PROB_c$). Ultimately, the objective function represents the patient's total waiting hours. The parameter $PROB_c$ reflects the contribution of each scenario to the overall solution, meaning the likelihood of each scenario occurring. Constraint (2) ensures that the total time a physician works during a week (the sum of the shift duration to which the physician has been assigned) does not exceed the allowed number of hours for the week. It only can be applied to physicians who work on weekdays. Constraint (3) is similar but applied to the weekend. Constraint (4)

guarantees that the total number of work hours allocated throughout the planning horizon does not surpass the permitted maximum hours. Constraint (5) dictates that the physician must be scheduled for back-to-back shifts, with only breaks in between. Constraint (6) ensures that each physician is allocated only up to the maximum number of shifts allowed for them on the same day. Constraint (7) ensures that the allocation of each physician adheres to the rules regarding the maximum consecutive working hours and avoids overlapping shifts by using the $REL_SHIFTS_{ss'}$ parameter, which indicates the relationship of shifts that can be combined due to these two factors. Constraint (8) permits a physician to work multiple shifts in a single day, provided that the shifts do not overlap, occur sequentially, and the physician belongs to the group authorized to work multiple shifts per day. Constraint (9) ensures that physicians working the night shift are not assigned any other shifts on the same day and can only work from the next night shift, if the parameter MAX_ASSIGN^1 is set to 1. If MAX_ASSIGN^1 is set to a value greater than 1, it will allow the allocation of the night shift with other shifts on the following day as many times as the value defined by the parameter. Constraint (10) ensures that each physician is assigned a maximum number of each type of shift throughout the planning period. Constraint (11) meets the minimum number of physicians per hour for each category. Constraint (12) enforces that the number of physicians allocated (by scheduling variable x_{sdp}) meets hourly the staffing levels defined for physicians (by staffing variable $nPhy_{htq}$). This constraint does not limit the number of physicians in each stage since the hospital of our case study has flexibility in allocating physicians in the workplace. Constraint (13) limits the number of patients served in the Triage phase according to the nurses' availability. Constraint (14) limits the number of patients served in the phases handled by the physicians. Constraint (15) specifies a bound for the number of patients served during a period. Constraints (16)–(17) limit the number of patients served in the Medication and Observation Hold according to the number of available chairs and beds, respectively. Constraint (18) limits the number of nurses available to serve the first queue. Constraint (19) guarantees the minimum number of physicians required in each queue. Constraints (20)–(24) update the number of patients in each queue at hours and days. In constraint (20) we observe the treatment of all patients that arrive in the ED (DEM_{htq}^c). After this stage, the patients are distributed following the transfer rates. The constraint (21) establishes that patients waiting in each hour in the First General Assessment Queue are patients waiting previously in this stage ($w_{h-1,t,q}^c$) adding patients from the triage phase ($\alpha_q s_{h-1,t,q-1}^c$), and discounting the patients served in the First General Assessment step ($s_{h-1,t,q}^c$). For the other phases of treatment, the number of patients waiting follows this same logic by changing only the parcels equivalent to the transfers of patients from different phases. Finally, constraints (25)–(29) define the domain of the decision variables.

4.2. Discrete-event simulation model

To evaluate the effectiveness of our optimization model, we compare the proposed optimal schedule and the current emergency department schedule using a stochastic discrete-event simulation model. The comparison includes metrics such as patient waiting times and frequency of queues. The simulation model developed in Arena® is based on the clinical pathway described in Section 3.2 and take the schedule and the number of physicians (per day and hour) specified by the optimization model as input.

Some assumptions taken for the mathematical model are relaxed in the simulation model: (i) patients requiring vital care that are seen in the Life-Threatening room are not treated just as a delay, being attended by a shared physician from the Observation Hold; (ii) all steps of treatment are no longer assumed to have a fixed duration but follow exponentially distributed inter-service times (as assumed in EL-Rifai et al., 2015); (iii) the system does not start empty, instead, we perform a warm-up period in the simulation model; (iv) the waiting time is

estimated precisely since the simulation uses a discrete event paradigm.

Moreover, we are considering a system with multiple available servers and accounting for variations in inter-arrival times daily and hourly. We conducted a Kruskal–Wallis test to evaluate whether the patient interarrival times were statistically significantly different on each day of the week and hour of the day. The current manually-defined ED schedule was used to define the warm-up period and number of required replications, and to validate our simulation model. Model validation was performed by comparing the actual number of patients treated in the ED with the number of patients attended in the simulation model.

5. Numerical experiments

We applied the solution framework to an actual case study in a Brazilian hospital. The scenario generation was implemented in MATLAB R2013a, and the mathematical programming model was implemented and solved using AIMMS 4.26 and GUROBI 10.0.0 with default settings. The solutions were achieved on an Intel i7 3.3 GHz 64 GB RAM computer. The simulation model was implemented using Arena Simulation software version 14.7. All statistical analyses were performed using R software version 3.3.1.

5.1. Case study

The patient flow is shown in Fig. 2, and the transfer rate of patients from one treatment stage to another is in Supplementary Material S4. After the triage, about 5% of the patients go straight to observation hold, 5% need vital care in the Life-threatening emergencies, and the others go to the first general assessment. Subsequently, to the first general assessment, 33.68% of the patients go on medication, 33.65% need auxiliary examinations, 10% go to the observation hold, and the others are discharged. Passing through medication, 25% of the patients still need auxiliary examinations. After the patient subsequent evaluation, 25% of the patients go to bed, 30% need medication, and 15% need other exams. Of patients needing treatment in a Life-threatening emergencies, 70% require auxiliary examinations.

The Hospital's Emergency Department (ED) information system provided 6,585 h of data from ED operations over ten months, including approximately 72,988 patient visits. The ED handles an average of 5,200 monthly medical visits for Clinical Medicine (CM), which are staffed by 85 physicians, with 1,457 work hours available for each planning horizon. General practitioners and cardiologists attend CM. Some physicians work as generalists and cardiologists, while others work exclusively as generalists or cardiologists. In total, 12 physicians can work as cardiologists. It is important to highlight that this hospital's Emergency Department operates 24 h a day, 365 days per year.

Physicians are assigned to one or more from 11 overlapping shifts (Fig. 3). The shifts differ in type according to the duration ($S_END_s - S_START_s + 1$) of each one, so the shifts can be of type 6 h, 9 h, 10 h, and 12 h. However, the 10 h shift (shift type 11) can only be used on weekends.

Physicians within the hospital are differentiated into two primary categories: permanent physicians and flexible physicians. Permanent physicians (14 physicians), who are considered the hospital's core medical staff, are assigned to a consistent weekly workload of 27 h during weekdays. These physicians are integral to the hospital's operations, ensuring stable and continuous care. On weekends, their commitment is adjusted to a single 10-h shift on both Saturday or Sunday, maintaining their presence within the hospital's scheduling framework. On the other hand, flexible physicians offer a more adaptable approach to scheduling. Unlike their permanent counterparts, flexible physicians have the liberty to set their availability on a monthly basis, offering between 12 to 24 h of service per week. This model caters to the diverse needs and availability of these physicians and the essence of the proposed model lies in its adaptability, designed to accommodate the specific

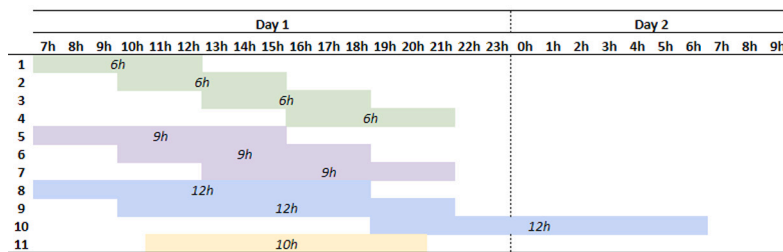


Fig. 3. ED shifts currently adopted by Clinical Medicine in Hospital A (24-h clock format).

needs and preferences of each physician. This approach ensures that the allocation of shifts is tailored to the individual, highlighting the model's capacity to adapt to various contexts and the unique characteristics of each medical professional.

There are also additional scheduling rules and agreements, including (i) combined allocation is allowed for non-overlapping and sequential shifts; (ii) A 4-week schedule must be created to ensure that both scheduling and staffing are consistent and cyclical. As a result, both the *PLAN_HORIZON* and the *PLAN_PERIOD* parameters are set to seven which means that decisions regarding both staffing and scheduling will be made for one week and repeated (in a cyclic manner) four times over the *PLAN_HORIZON* (as detailed in the Supplementary Material S3); (iii) there should be a minimum of four physicians allocated at all times, with at least one being a cardiologist.

To determine service capacity, we adopted the following assumptions (corroborated by ED physicians and managers): each nurse in the triage can see eight patients per hour (the average service time is 7.5 min); physicians in the first and subsequent evaluations can see three patients per hour (i.e., the average service time is 20 min); physicians in the medication can treat six patients simultaneously; and physicians in the observation hold can treat four patients simultaneously.

Currently, staffing at the ED is performed by the ED Administration through an empirical analysis of demand behavior. Generating a manually-defined schedule is laborious, and its solution is not necessarily optimal. Although the current schedule (shown in Supplementary Material S5) follows all scheduling rules and agreements, it does not consider the demand fluctuation which generates a gap between the current demand level and ED capacity.

The hospital administration suggests that the different shift types of ED (6 h, 9 h, 10 h, and 12 h) would be allowed to start at other times of the day (as shown in Fig. 4) to increase the number of shifts for physicians, but this is not implemented due to the difficulty of manually creating a schedule with so many possibilities. It is worth noting that the 10-h shifts (shifts 26–30) can only be utilized on weekends. We will refer to this scenario of shift distribution as the “Proposed Shift Distribution - PROP”, since it is a scenario desired by the hospital and is only not implemented due to difficulties in manually definition of the allocation in these shifts. The case study with the original data does not account for breaks in shifts. In the results section, we will take into account the breaks as part of the model's sensitivity analysis.

5.2. Model implementation and results

Patient arrival rates at “Hospital A” were modeled as Poisson processes. This distribution was selected to best fit the actual patient arrival data (chi-square test p -value of 0.216). We used this probability distribution to perform the random sample draws in the generation of scenarios regarding the arrival of patients (demand). The number of scenarios employed in the stochastic optimization using SAA is calculated according to the following steps: (1) for lower bound calculation, in our in-sample analysis, we use ten replications ($M = 10$), and we start using 20 scenarios ($N = 20$); (2) we use an additional independent set of samples of 1,000 independent scenarios ($T = 1$ and $N = 1,000$),

Table 5

Size of SAA's experiments.

Scenarios	Variables	Constraints	Time(s)	St.Dev(s)
20	57 835	351 553	671.66	115.29
40	98 155	411 893	1608.75	514.95
60	138 475	472 233	2510.03	474.17
80	178 795	532 573	3291.81	281.66
100	219 115	592 913	3541.32	0.24

Table 6

Experiment results: estimative of the optimality gap and confidence interval.

Scenarios	%Gap	Confidence interval	
		Min	Max
20	5.157	4.711	5.603
40	2.324	2.028	2.620
60	1.995	1.712	2.274
80	0.995	0.829	1.171
100	0.011	−0.044	0.067

representing the true distribution, to calculate the upper bound, our out-of-sample analysis (3) if the stop criterion (the difference of 1% between the lower and upper bounds, i.e. an estimated gap of 1% with 95% level of confidence interval) is not achieved, the required number of scenarios N is increased by 20. The stopping criterion for each replication was 3,600 s computational time.

In this Section, we aim to compare the results of four scenarios: (i) Manual Allocation (MANU): current manually-defined ED schedule that uses the shift distribution as depicted in Fig. 3; (ii) Model with Current Shift Distribution (CURR): which utilizes the same shift combination currently used by the hospital; (iii) Model with Proposed Shift Distribution (PROP): which utilizes a combination of shifts shown in Fig. 4, considering possible shifts distribution for hospital application; and (iv) Model with Fully Flexible Shift Distribution (FLEX) explores a completely flexible shift scenario (refer to Supplementary Material S6 for the shift details). It is important to note that this last scenario is designed solely to gauge the potential benefits of shift flexibility. However, it is not considered viable for implementation in Hospital A's operations.

Before we run the results for the cases presented earlier, we need to define the number of scenarios that should be used in the experiments. For this, we will use the “PROP” case as a basis. We run preliminary tests using Latin Hypercube Sampling (LHS). The size of the models (i.e., variables and constraints), average execution times, and their variances are shown in Table 5. The average solution time of the experiments was 676.36 CPU seconds for 20 scenarios and 3,602.60s for 100 scenarios. Table 6 presents the results for each experiment in terms of the estimate of the optimal gap to the true solution, showing the confidence interval.

The gap values vary from 5.163% in the experiments with 20 scenarios to 0.014% in the experiments with 100 scenarios. Therefore, we notice that the optimal gap achieves the stop criterion defined for 100

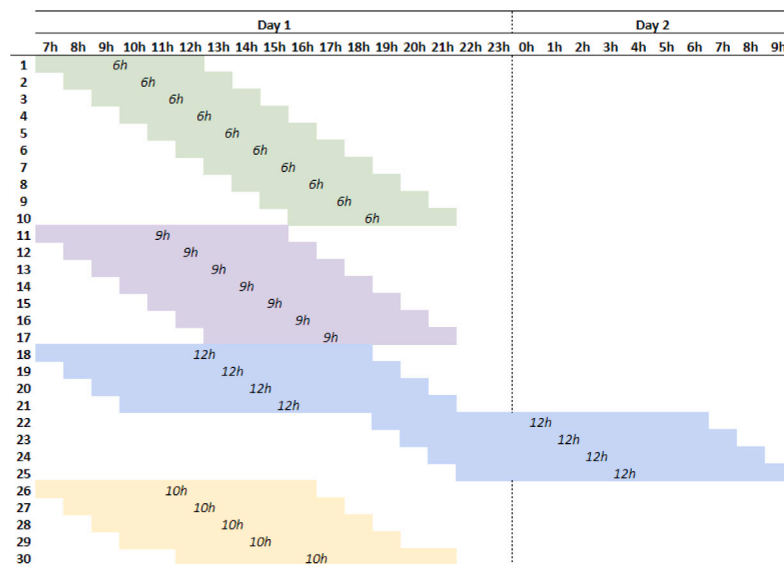


Fig. 4. Proposed Shift Distribution to use in Hospital A's ED (24-h clock format).

scenarios, being the 95% level of confidence interval (CI) 0.056; thus, the gap considering the CI is lower than 1%. So all results presented hereafter will be using this scenario setting.

After defining the ideal number of scenarios, we ran the optimization model for each of the cases previously presented in order to define the optimal staffing and scheduling. Then, we used the discrete-event simulation model to estimate the benefits of our model's results compared to the current staffing and scheduling. For the simulation model of the studied ED, we assumed the premises cited in Section 4.2. The Kruskal–Wallis test revealed a statistically significant difference in patient inter-arrival times in each day of the week and hour of the day (chi-squared = 8083.6, $df = 23$, p -value < $2.2e-16$). We used the Kruskal–Wallis test due to the non-normality of the data set (Shapiro test, $W = 0.7866$, p -value < $2.2e-16$). Then, we performed a multiple comparison test to define which hours and days of the week could be considered to have the same arrival rate and, hence, be grouped. The Supplementary Material S7 shows the average interval between arrival per hour of the day and day of the week in Hospital A. Hours with no statistically significant arrival difference are presented in the same group.

We employed a 12-h warm-up period and conducted 100 replications to ensure that system occupancy accurately reflects real-world initial conditions. Each replication simulated 6,585 h of operations, which amounts to the total length of the period for which ED data was made available. Model validation was performed by comparing the real number of patients treated in the ED (80,287 patients) with those attended in the simulation model (80,051 patients), yielding a variation of approximately 0.29% in the total number of treated patients. Additionally, two key indicators, Length of Stay (LOS) and Overall Waiting Time, were assessed. The actual average LOS was 102.06 min, whereas the average LOS in the simulation model was 101.55 min. Similarly, the real Overall Waiting Time was 54.56 min, while it was recorded as 53.89 min in the simulation model. We considered such variation negligible, so the simulation model was validated.

Results of the numerical experiments are presented in Table 7. In addition to the value of each indicator, we also report the 95% level of confidence interval (CI). The physician allocation resulting from the PROP is shown in Supplementary Material S8. Based on the detailed results observed in the Table 7, we delve into the comparative analysis of different staffing and scheduling scenarios in a hospital setting. The study contrasts the current manual allocation of hospital shifts, referred to as “MANU”, with three model-based allocations as explained above.

The CURR scenario, even without the flexibility of shifts, showcases an improvement over the current manual system by better aligning physician allocations with demand. This is evidenced by the indicators such as the frequency of queue in the First Assessment phase dropping from 31.14% in the current allocation to 21.50%, and the LOS reducing from 102.06 min to 94.95 min. Such improvements underscore the model's capability to enhance queue indicators significantly.

The PROP scenario is highlighted as the study's proposed case for adoption over the current state. This case, by incorporating a level of shift flexibility, further optimizes outcomes. For instance, in the first assessment phase, the frequency of queue is markedly reduced to 12.16%, and the LOS is cut down to 64.33 min. These results underscore the advantages and feasibility of transitioning from manual to model-based scheduling. Notably, the model's ability to automate the allocation process enables the hospital to adopt more complex scheduling scenarios without the burden of manual intervention. The increased complexity, far from being a drawback, is actually an asset, as it allows for a more nuanced and efficient allocation that would be impractical to achieve manually.

These findings highlight the importance of accommodating variations in physician entry times to optimize resource allocation and streamline operations. This can be better observed when we compare the usage of different shifts throughout the week and the treatment stages of the MANU scenario with the PROP scenario (see Supplementary Material S9). Upon comparing the frequency of usage of each shift, we find that the 6-h and 12-h shifts are still the most utilized, similar to the manual allocation implemented by the hospital. However, in the PROP scenario, the model explores the flexibility of the start times of the 6-h and 9-h shifts more extensively. This is because the model optimizes shift coverage throughout the day, ensuring sufficient staff at all times when demand is high. Furthermore, we can observe that this flexibility is exploited even more in the stages of First and Review Assessments, as these stages have a higher volume of patients and require longer service times, needing better physician scheduling. Another important point regarding allocation and shift selection is that cardiologists (who are less available) are allocated to longer shifts, aiming for greater coverage of this category throughout the day.

Lastly, the FLEX scenario, while not deemed viable for practical implementation at the hospital where the study was conducted, is presented to illustrate the upper bounds of shift flexibility benefits. The data reveal that even more substantial improvements could be realized if full flexibility were possible, with the overall average waiting time

Table 7
Discrete-event simulation results.

Treatment phase	Indicators	MANU		PROP		CURR		FLEX	
		Value	CI	Value	CI	Value	CI	Value	CI
First assessment	Frequency of queue (%)	31.14	0.00	12.16	0.00	21.50	0.00	15.90	0.00
	Queue above 10 patients (%)	11.02	0.00	1.40	0.00	4.71	0.00	2.72	0.00
	Queue between 1 and 10 (%)	20.12	0.00	10.46	0.00	16.5	0.00	12.87	0.00
	Average number in queue	3.05	0.05	0.59	0.01	1.51	0.01	0.96	0.01
	Average waiting time (min)	36.55	0.54	7.17	0.16	18.14	0.62	11.56	0.12
	Average scheduled utilization (%)	51.6	0.00	41.59	0.00	57.07	0.00	47.78	0.00
Medication	Frequency of queue (%)	5.58	0.00	10.38	0.00	12.05	0.00	11.05	0.00
	Queue above 10 patients (%)	0.18	0.00	0.29	0.00	0.34	0.00	0.31	0.00
	Queue between 1 and 10 (%)	5.4	0.00	10.10	0.00	11.72	0.00	10.75	0.00
	Average number in queue	0.18	0.01	0.33	0.01	0.38	0.01	0.35	0.01
	Average waiting time (min)	4.46	0.22	7.94	0.14	9.28	0.25	8.48	0.08
	Average scheduled utilization (%)	30.71	0.00	38.40	0.00	47.61	0.00	42.09	0.00
Subsequent evaluation	Frequency of queue (%)	45.48	0.00	21.09	0.00	34.73	0.00	26.55	0.00
	Queue above 10 patients (%)	6.71	0.00	0.99	0.00	3.00	0.00	1.80	0.00
	Queue between 1 and 10 (%)	38.76	0.00	20.10	0.00	31.73	0.00	24.75	0.00
	Average number in queue	2.52	0.02	0.74	0.01	1.5	0.01	1.04	0.01
	Average waiting time (min)	55.74	0.50	16.52	0.24	33.24	0.53	23.21	0.17
	Average scheduled utilization (%)	65.35	0.00	47.12	0.00	66.73	0.00	54.96	0.00
Observation	Frequency of queue (%)	1.74	0.00	5.08	0.00	5.6	0.00	5.29	0.00
	Queue above 10 patients (%)	1e−7	0.00	0.01	0.00	0.01	0.00	0.01	0.00
	Queue between 1 and 10 (%)	1.74	0.00	5.06	0.00	5.58	0.00	5.27	0.00
	Average number in queue	0.04	0.00	0.11	0.00	0.12	0.00	0.11	0.00
	Average waiting time (min)	1.26	0.05	3.89	0.11	4.27	0.09	4.04	0.15
	Average scheduled utilization (%)	24.86	0.00	36.20	0.00	43.66	0.00	39.18	0.00
Overall average waiting time (min)		54.56	0.52	16.83	0.18	33.2	0.57	23.38	0.16
Length of stay (min)		102.06	0.33	64.33	0.15	94.95	0.34	44.56	0.13

¹ MANU - Manually-defined ED schedule; PROP - Model with Proposed Shift Distribution; CURR - Model with Current Shift Distribution; FLEX - Model with Fully Flexible Shift Distribution; CI - Half-width confidence interval.

across all phases decreasing to 23.38 min from 54.56 min in the current allocation. This scenario, albeit hypothetical, serves to highlight the model's comprehensive approach, addressing staffing and scheduling in unison, accommodating multiple stages of emergency care, and integrating physician preferences with both cyclic and acyclic shift periodicity options.

In conclusion, the study's findings suggest that a model-based approach to hospital shift allocation can significantly improve service levels by optimizing across multiple treatment stages. This not only enhances the immediate patient care stages but also demonstrates the potential for a global optimization of service levels, where improvements in one stage can be achieved with minimal impact on the flexibility of others. "PROP" cases emerges as a particularly compelling alternative to current practices, promising substantial improvements in efficiency and patient wait times without the need for complete shift flexibility.

5.3. Model's sensitivity analysis

To enhance the robustness of the schedules generated across different operational scenarios, we conducted a sensitivity analysis. Initially, we varied service times while keeping the original number of physicians constant. Our investigation encompassed four distinct scenarios: Scenario 1%–25% increase in all service times; Scenario 2%–50% increase in all service times; Scenario 3%–25% reduction in all service times; Scenario 4%–50% reduction in all service times. The detailed service time information for each scenario are presented in Supplementary Material S10.

Subsequently, we delved into variations in the number of available physicians, encompassing both general practitioners and cardiologists. The ensuing scenarios are as follows: Scenario 5 - Approximately 5% reduction to General Practitioners (12 cardiologists and 69 General Practitioners); Scenario 6 - Approximately 4% reduction to General

Practitioners and 8% reduction to cardiologists (11 cardiologists and 70 General Practitioners) representing 5% reduction in the total; Scenario 7 - Approximately 12% reduction to General Practitioners (12 cardiologists and 64 General Practitioners) representing 10% reduction in the total; Scenario 8 - Approximately 9% reduction to General Practitioners and 17% reduction to cardiologists (10 cardiologists and 66 General Practitioners) representing 10% reduction in the total; Scenario 9 - Approximately 18% reduction to General Practitioners (12 cardiologists and 60 General Practitioners) representing 15% reduction in the total; Scenario 10 - Approximately 14% reduction to General Practitioners and 25% reduction to cardiologists (9 cardiologists and 63 General Practitioners) representing 15% reduction in the total. For each of these ten scenarios, we can make a correspondence with real-world situations (as detailed in Supplementary Material S10).

Another sensitivity analysis involved the inclusion of breaks within the considered shifts. For all shifts, one-hour breaks were incorporated at the midpoint of each shift (details on how the shifts were structured with breaks can be found in Supplementary Material S11). We consider these changes as the scenario 11 of the sensitivity analysis. We also ran the "MANU" scenario considering breaks within the shifts to enable a fairer comparison with the outcome proposed by the model. The summary of the results from the sensitivity analysis is presented in Table 8. Detailed results for each scenario within the sensitivity analysis are comprehensively detailed in Supplementary Material S12.

Based on the sensitivity analysis of service time and number of physicians, the proposed model exhibits robust performance across various scenarios compared to the current allocation. In scenarios 1 and 2, the proposed model consistently outperforms the current allocation in terms of queue frequency, average queue, and average waiting time in the first assessment and subsequent evaluation phases. This trend is also observed in scenarios 3 and 4, where there is a reduction in all service times by 25% and 50%, respectively. When the number of physicians is reduced, as in scenarios 5 to 10, the proposed model

Table 8
Sensitivity analysis' summary results.

Analyze type	Scenarios	Overall average waiting time (min)	Length of stay (min)
Baselines Scenarios	MANU	54.56	102.06
	PROP	16.83	64.33
Service time variation	Scenario 1	24.47	46.60
	Scenario 2	26.11	49.66
	Scenario 3	15.47	29.76
	Scenario 4	14.1	27.21
Physician's number variation	Scenario 5	21.19	68.69
	Scenario 6	22.83	70.33
	Scenario 7	23.92	71.42
	Scenario 8	25.56	73.06
	Scenario 9	26.65	74.15
	Scenario 10	28.29	75.79
Breaks' Consideration	MANU w/ breaks	68.09	126.35
	Scenario 11	30.47	89.85

performs better than the current allocation, albeit with a slight increase in queue frequency and waiting time.

The sensitivity analysis results regarding break consideration indicate that introducing breaks negatively impacts LOS performance. Nevertheless, the proposed model with breaks (scenario 11) achieves better results than the “MANU” scenario with breaks. The LOS achieved with scenario 11 is 29% lower than the “MANU” with breaks. Compared to the “PROP” scenario, the result of scenario 11 shows a 40% higher LOS. The behavior of queues and time between different treatment stages is similar to that observed in other sensitivity analysis scenarios. Compared to other scenarios, this scenario presents a higher LOS than all previous scenarios.

It is important to note that our model considers flexibility in shift start times and incorporates individual physician preferences and restrictions regarding the start time in the ED. The combination of intricate shift configurations, contractual agreements, and service requirements necessitates the integration of staffing and scheduling phases. Our study is the first to integrate staffing and scheduling across multiple treatment stages within an Emergency Department (ED) setting. The results indicate that the schedule generated by the proposed model achieves better results than the current manually-defined ED schedule.

6. Final considerations

The present article proposed a generalized and integrated model that considers uncertainty in patient arrival patterns and simultaneously accounts for ED physician staffing and scheduling problems. We applied a two-stage stochastic approach involving scenarios using the SAA and analyzed patient arrival uncertainty throughout discrete event simulation.

We applied the model in an actual case study with 10-month period data, including approximately 72,988 patient visits and staffed with 85 physicians. The simulation results, including the uncertainty in patient arrival, showed that the optimal schedule leads to reductions of more than 50% in the frequency of queues for most ED stages, showing that the model better distributed the doctors between the areas to reduce the total waiting time (overall average waiting time is reduced from 55 to 17 min). Therefore, this work proposed a stochastic model that effectively accommodates uncertainties, identifying solutions to a relevant healthcare problem and including stochastic aspects of the system.

While our study has limitations, mainly due to the dataset used, we believe these do not hinder our model's applicability. The dataset provides service time information for each treatment, but lacks stratified data for individual physicians and nurses. However, our model can accommodate different service times for each specific physician and nurse.

By implementing the solution framework, we achieve a notable alignment between service capacity and demand, resulting in a substantial enhancement in the frequency of queue and waiting time indicators.

Furthermore, it is crucial to emphasize that the optimization approach proposed in this study can be applied to other Emergency Department (ED) settings. Our study assumes consistent physician service rates throughout the planning period, ensuring urgent care patients receive prompt screening and treatment. However, it is essential to incorporate variations in patient acuity and fluctuating service times, which are significant in actual emergency departments. Additional developments of this work should adapt the model for vacation schedules, different service rates for each physician category and explore the model's selection of shifts while maintaining certain fixed doctor allocations.

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CRedit authorship contribution statement

Janaina F. Marchesi: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Silvio Hamacher:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Igor Tona Peres:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT to improve readability and language. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ejor.2025.01.027>.

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